

Biology by the Numbers Spring 2026: Immunology by the Numbers — the Antibody Repertoire

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How does the body recognise *antigens*? The word itself is a giveaway: *anti(body)-gen(erator)* — anything that triggers an antibody response. Pathogen surfaces, toxins, drugs, DNA, metals, polymers, sugars, lipids, self-molecules, and even other antibodies. The list goes on.

Antibodies are distinct from other proteins: their binding site is rearranged at the DNA level. This is done by an idiosyncratic mechanism — V(D)J recombination — which assembles a unique receptor in each developing lymphocyte and commits every B cell to one specific binding site shape out of *quadrillions* of possible alternatives. By producing hundreds of millions of B cells, the body guarantees coverage of a vast space of shapes, like a spider's web that vibrates the instant a prey lands anywhere on it. Recognition of an antigen sets off a wave of clonal expansion that amplifies neutralization and downstream effector functions.

This lecture explores how big the antibody repertoire is, how it is assembled, where its diversity comes from, and how the system learns to protect us from a second encounter — the phenomenon we call *immunity*.

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1 Foundations: where antibodies come from

1.1 B-cell development

All blood cells originate from haematopoietic stem cells (HSCs) in the bone marrow. The first major branch separates the myeloid lineage (granulocytes, monocytes, megakaryocytes) from the lymphoid lineage. Within the lymphoid branch, the *common lymphoid progenitor (CLP)* gives

HSC pool. Adult humans maintain $\sim 10^4$ – 10^5 actively contributing HSCs [Lee-Six 2018; Catlin 2011], collectively producing $\sim 5 \times 10^{11}$ blood cells per day [Taichman 2005; BNID 113199].

Lymphocyte output. Bone marrow generates $\sim 10^7$ – 10^8 new B cells daily [Macallan 2005]. In adult humans the thymus contributes only ~ 10 – 20% of naive T-cell production [Den Braber 2012; BNID 113830] — the rest comes from peripheral homeostatic proliferation. Productive thymus decays with a ~ 16 -year half-life [Murray 2003], $\sim 4\%$ per year.

rise to B cells, T cells, and NK cells.

B and T cells share this common progenitor; what distinguishes them is anatomical:

B cells complete maturation in the bone marrow.

T-cell precursors migrate to the thymus to mature there.

	Mouse	Human
Naive B-cell pool	$\sim 2 \times 10^8$	$\sim 4 \times 10^{11}$
B-cell production /day	$\sim 10^7$	$\sim 10^7$ – 10^8
Total body cells	$\sim 2 \times 10^{10}$	$\sim 3 \times 10^{13}$
B-cell / body fraction	$\sim 1\%$	$\sim 1\%$
Daily turnover	~ 5 – $10\%/d$	~ 0.003 – $0.025\%/d$

On the names. “T” = thymus. “B” originally meant *Bursa of Fabricius*, the avian organ where Glick (1956) first linked an organ to antibody production — not bone marrow. Bone marrow only later turned out to be the mammalian equivalent; the letter survived.

Table 1: Mouse and human B-cell numbers.

1.2 *V(D)J recombination*

Each B cell carries a unique BCR assembled by cutting and pasting gene segments at the immunoglobulin loci. The recombination is final: each B cell is committed to one antibody specificity for its lifetime.

1. **RAG1+RAG2 cleaves the DNA.** A heterotetramer (RAG1₂RAG2₂). RAG1 carries the catalytic core; RAG2 reads H₃K₄me₃ (an active-chromatin mark) and acts as a developmental clock — it is degraded at the G₁/S transition, restricting recombination to G₁ phase.
2. **V(D)J is somatic DNA editing.** Permanent and inherited by all clonal descendants. Unlike RNA splicing (reversible, per-transcript), V(D)J cuts and rejoins the chromosome itself. *One cell, one rearrangement.*
3. **Both B and T cells use V(D)J recombination** (see §1.1) but deploy the receptor differently (next).
4. **Effector mode differs.** T cells keep the receptor anchored to the membrane and act through contact. B cells secrete the receptor — the antibody is, in effect, the BCR set free — and broadcast specificity into the fluid phase.

1.3 *The immunoglobulin loci*

In humans the IgH locus is on chromosome 14 and contains ~ 50 functional V_H segments, ~ 27 D segments, and 6 J_H segments, followed by the constant-region exons (C_μ, C_δ, C_γ, C_α, C_ε). The Igκ locus is on chromosome 2 and Igλ on chromosome 22, each with V and J segments and a single constant region per chain. Heavy and light chains are recombined in sequence: heavy first (to make a pre-BCR), then light.

Recombination signal sequences (RSSs) flank each segment; RAG cuts between segment and RSS. The 12/23 rule (one 12-bp-spaced and one 23-bp-spaced RSS per join) prevents V–V or J–J fusions.

Successful expression of both produces a complete IgM BCR; failure at any step deletes that cell.

For practical purposes the selection of V, D, and J segments can be treated as random, but in reality each segment has a different intrinsic propensity to be targeted by RAG — driven by 3D locus geometry, RAG scanning, RSS strength, and proximity to J_H . As a result, V-gene usage in the realised repertoire is heavily biased; we will return to this in §5.

2 Three guiding questions

Three questions structure the rest of the lecture.

Q1. *How diverse does the antibody repertoire need to be for competent humoral immunity?*

Q2. *Why would evolution maintain such an expensive mechanism unless it carries a clear fitness benefit?*

Q3. *If more diversity is better, why do repertoire sizes vary so much across species?*

3 The demand side — how big does the repertoire have to be?

The number of *distinct epitopes* an animal's immune system might have to recognise is the demand-side ceiling on repertoire size. A back-of-envelope estimate (Rees 2020, *mAbs*):

$$\begin{aligned} \text{epitope universe} &\approx \underbrace{1,400}_{\text{human pathogens}} \times \underbrace{10}_{\text{antigens per pathogen}} \times \underbrace{100}_{\text{epitopes per antigen}} \\ &\approx 1.4 \times 10^6 \text{ epitopes.} \end{aligned}$$

So the demand-side answer to Q1 is $\sim 10^6$ distinct antibody specificities.

The Perelson coincidence. Perelson and Oster (1979) reached the same $\sim 10^6$ from a completely independent argument: treat antibody binding as covering a high-dimensional “shape space,” and ask for the smallest N_{Ab} that reliably covers it. Two unrelated paths (counting pathogens, covering shape space) converge on the same minimum.

The counterfactual is striking. Humans with non-functional V(D)J (SCID, RAG1/RAG2 mutations, Artemis deficiency, Omenn syndrome) die in infancy without intervention. And jawless vertebrates (lampreys, hagfish) independently evolved a separate adaptive system using *variable lymphocyte receptors* built from leucine-rich repeats (Pancer & Cooper). Two independent inventions of adaptive immunity in vertebrates argue strongly for the fitness benefit.

Pathogen count from Taylor, Latham & Woolhouse 2001 *Phil Trans R Soc B*: 1,415 species pathogenic to humans (217 viruses, 538 bacteria, 307 fungi, 66 protozoa, 287 helminths). The other two factors are admitted order-of-magnitude estimates.

Read more: Perelson & Oster 1979, *J. theor. Biol.* 81, 645–670 — the original shape-space derivation.

4 The supply side — how many antibodies can $V(D)J$ produce?

The theoretical repertoire combines gene-segment combinatorics, junctional editing (resection + P + N nucleotides), and heavy/light pairing into a single product:

$$R_{\text{theo}} = \underbrace{(V_H \cdot D \cdot J_H)}_{\text{H combinatorial}} \cdot \underbrace{(n_V+1)(n_D+1)(n_J+1)}_{\text{H resection}} \cdot \underbrace{P_H \cdot 4^{N_H}}_{\text{H palindromic + untemplated}} \times \underbrace{[(V_\kappa J_\kappa) + (V_\lambda J_\lambda)]}_{\text{L combinatorial}} \cdot \underbrace{(n_V+1)(n_J+1)}_{\text{L resection}} \cdot \underbrace{P_L \cdot 4^{N_L}}_{\text{L palindromic + untemplated}}. \quad (1)$$

Terms. V, D, J are functional gene-segment counts. n_X is the maximum number of nucleotides exonucleases can resect off segment X (each resection length $0 \dots n$ contributes one option, hence $n+1$). P is the P-nucleotide (palindromic, templated) factor. N is the average count of TdT-added non-templated nucleotides at the join, each chosen from the 4 bases $\{A, T, G, C\}$ (hence 4^N). Subscripts H, L denote heavy and light chains.

We now work the mouse case stage-by-stage, then compare against human and cat.

4.1 Mouse, human, and cat segment counts

We will work the mouse case in full below; first a side-by-side comparison of three species reveals strikingly different lineage strategies for distributing diversity between heavy and light loci.

Species	V_H	D	J_H	V_κ	J_κ	V_λ	J_λ
Mouse (C57BL/6)	~ 100	~ 12	4	~ 100	4	~ 3	3
Rat	~ 100	$\sim 10^+$	~ 4	~ 135	5	few	3
Human	~ 46	~ 23	6	~ 38	5	~ 30	4
Rabbit	~ 14	$\sim 10^+$	6	~ 20	5	~ 20	~ 6
Cow	~ 12	$\sim 10^+$	6	few	5	~ 20	9
Cat	~ 42	$\sim 5^+$	$\sim 4^+$	~ 18	5	~ 94	~ 12
Chicken	1	4	1	—	—	1	1

Table 2: Functional Ig gene-segment counts across species (IMGT; order-of-magnitude estimates where IMGT curation is incomplete). Daggers ($^+$) flag values inferred from genome-assembly annotation rather than full IMGT curation. Chicken uses one functional V_H and one functional V_L alongside ~ 95 V-pseudogenes per locus, repurposed as gene-conversion donors in the bursa.

4.2 Mouse C57BL/6 worked example

Step 1 — Combinatorial.

$$\text{IgH}_n = 100 \cdot 12 \cdot 4 \approx 5 \times 10^3, \quad \text{IgL}_n = (100 \cdot 4) + (3 \cdot 3) \approx 4 \times 10^2.$$

Mouse is κ -dominant ($\sim 95\%$). TdT activity is minimal during light-chain rearrangement, so N-additions are essentially absent on the light chain — which is why the light-chain junctional factor is small.

Step 2 — Junctional. Typical mouse values: per-segment resection $n \approx 3$ for each segment, P-nt factor $P_H \approx 5$, N-additions $N_H \approx 4$. Light chain has resection only ($n \approx 3$, no P, no N):

$$\begin{aligned} \text{res}_H &= (3+1)^3 = 64, & \text{res}_L &= (3+1)^2 = 16, \\ P_H &= 5, & P_L &= 1, \\ 4^{N_H} &= 4^4 = 256, & 4^{N_L} &= 1. \end{aligned}$$

Step 3 — Per-chain total.

$$\begin{aligned} \text{IgH}_{\text{total}} &= 5 \times 10^3 \cdot 64 \cdot 5 \cdot 256 \approx 4 \times 10^8, \\ \text{IgL}_{\text{total}} &= 4 \times 10^2 \cdot 16 \cdot 1 \cdot 1 \approx 6 \times 10^3. \end{aligned}$$

Step 4 — Pairing.

$$R_{\text{theo}}^{\text{mouse}} = \text{IgH}_{\text{total}} \cdot \text{IgL}_{\text{total}} \approx 4 \times 10^8 \cdot 6 \times 10^3 \approx 2 \times 10^{12}.$$

4.3 Brief comparison of species totals

Plugging segment counts from Table 2 into equation (1) with broadly mammalian junctional parameters (same res , P , N values as mouse, except human has detectable light-chain TdT activity) gives the following V(D)J-only estimates:

Species	R_{theo}	Dominant mechanism
Mouse (C57BL/6)	$\sim 2 \times 10^{12}$	V(D)J + TdT
Rat	$\sim 3 \times 10^{12}$	V(D)J + TdT
Human	$\sim 10^{14}$	V(D)J + TdT
Cat	$\sim 10^{12}$	V(D)J + TdT
Rabbit	$\sim 10^{11}$	V(D)J + gene conversion (appendix)
Cow	$\gtrsim 10^{20}$	V(D)J + ultralong CDR3 (>50 aa)
<i>Outside the V(D)J formula:</i>		
Chicken	$\sim 10^8 \ddagger$	gene conversion only (~ 95 pseudo-donors)

$\ddagger$ Chicken has only 1 functional V_H and 1 functional V_L ; eq. (1) yields $\sim 10^2$ for chicken. The realised repertoire is generated by a separate post-rearrangement mechanism (gene conversion in the bursa) that is not part of the formula.

Table 3: Theoretical repertoire estimates by species. Rows above the horizontal rule are well described by eq. (1): cow simply uses a much larger N , and rabbit's gene conversion is layered on top of a normal V(D)J base. Chicken (below the rule) sits in a different regime: V(D)J alone yields only $\sim 10^2$, and the realised repertoire comes entirely from gene conversion off ~ 95 pseudogene donors — eq. (1) does not model this mechanism.

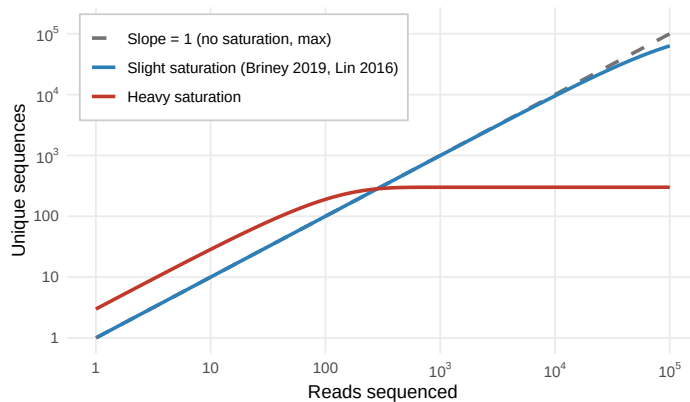
Three lessons. First, the standard-mammal species (mouse, rat, human, cat) all reach $\sim 10^{12}$ – 10^{14} through V(D)J + TdT, distributing diversity very differently between loci: human gains over mouse via balanced κ/λ usage and a larger D pool; cat compresses its heavy locus but compensates with a λ expansion (94 V_λ). Second, cow climbs *dramatically above* the standard-mammal estimate by exploiting a single very long CDR3_H (>50 aa). With $N \sim 30$ non-templated nucleotides added at the join, the 4^N term alone is $\sim 10^{18}$; multiplied by combinatorial (and possibly extra D-D fusions) and heavy/light pairing the total exceeds

10^{20} . Equation (1) still applies — cow just turns the N dial much higher than mammals.

Third, chicken sits in a different regime entirely. Its IGH locus has a single functional V_H and a single functional J_H , and the light-chain locus is just as collapsed. Plugging into eq. (1) gives $\sim 10^2$ — nowhere near enough. The realised repertoire of $\sim 10^8$ – 10^9 unique BCRs comes from a fundamentally different mechanism: *gene conversion* in the bursa of Fabricius, where ~ 95 V-pseudogenes per locus serve as templates that copy short tracts into the single rearranged V. This is post-recombination editing, not V(D)J + junctional editing — our formula simply does not describe it. Chicken is the cleanest reminder that V(D)J is one solution to the diversity problem, not the only one.

4.4 Sampled vs. realised vs. theoretical

A subtle point: the numbers people quote for “realised” repertoire come from **deep sequencing** of B cells in blood. They are *sampled*, not realised — and they substantially undercount the true diversity because the rarefaction curves have not plateaued. Each additional read still discovers new unique sequences.



Briney et al. 2019 sequenced ~ 3 billion BCR reads from human donors but recaptured only 3.9–11.7% of unique sequences across replicates. Their extrapolated cohort-wide nucleotide diversity is $\sim 10^{15}$ (heavy chain only) and $\sim 10^{16}$ – 10^{18} paired. Lin 2016 in mouse has the same problem: 10^7 is what they observed, not what the animal carries.

Rarefaction curve schematic (generated by R/rarefaction_plot.R). Cumulative unique

BCRs as a function of reads sequenced (log–log). The dashed slope-1 line is the maximum possible: every read a brand-new sequence. Real curves lie at or below it. **Blue** = slight saturation (Briney 2019, Lin 2016) — huge true diversity remains. **Red** = heavy saturation — nearly all diversity captured. The sequenced counts are therefore *lower bounds*.

So the right bracket is three-tier:

Source	N_{Ab}
Demand (Rees, Perelson)	$\sim 10^6$
Sampled (Lin 2016, mouse; Briney 2019, human)	$10^7\text{--}10^9$
Estimated realised (rarefaction-extrapolated)	$\gtrsim 10^{11}\text{--}10^{12}$
Theoretical (full V(D)J)	$\sim 10^{12}\text{--}10^{14}$

The “realised” repertoire is much closer to the V(D)J theoretical ceiling than the sequencing numbers naively suggest. The system carries more than enough diversity to cover the demand; the apparent gap between 10^7 and 10^{12} is partly an artefact of incomplete sampling. The gap between realised and theoretical motivates the next section.

5 *Why realised $\ll$ theoretical: where reality departs from the formula*

Equation (1) is built on four simplifying assumptions. Each holds approximately, but reality refines them in ways that pull the realised repertoire down by $\sim 10^2\text{--}10^3$ from the theoretical ceiling. The point isn’t that the formula is wrong — it correctly counts what the machinery can in principle assemble. The realised repertoire is a *biased sample* of that space, shaped by chromosome architecture, developmental regulation, and pairing constraints.

Assumption in eq. (1)	How biology refines it
Every V(D)J resection is equally likely.	The exonuclease that trims segment ends is sequence-context-dependent; some end sequences are preferred [Murugan 2012; Elhanati 2015].
Every junction has P and N nucleotides.	TdT is developmentally regulated; fetal and neonatal repertoires lack N-additions almost entirely [Komori 1993; Gilfillan 1993].
Every V gene is equally likely to recombine.	3D locus contraction, RAG scanning, RSS strength, and proximity to J_H all bias which V genes are reached [Larimore 2012; Dai 2021; Greiff 2017].
Every $V_H\text{--}V_L$ pair is possible.	Heavy and light chains show <i>light-chain coherence</i> : some pairings are favoured and others systematically avoided [Jaffe 2022 <i>Nature</i>].

6 *Are the contacts tight enough? — K_d and epitope binding*

Sections 3–5 said the repertoire is *big enough* to cover antigenic space. This section asks the next question: are the individual antibody–epitope

contacts *tight enough* to actually neutralise pathogens at realistic concentrations?

6.1 Equilibrium binding — the back-of-envelope

Single antibody binding a single epitope:

$$\text{Ab} + \text{Ep} \rightleftharpoons \text{Ab} \cdot \text{Ep}, \quad K_d = \frac{k_{\text{off}}}{k_{\text{on}}} = \frac{[\text{Ab}][\text{Ep}]}{[\text{Ab} \cdot \text{Ep}]}.$$

Bigger K_d means weaker binding. Under the realistic regime where specific antibody is in excess over the target epitope (true in serum vs. pathogen), the fraction of epitope bound is

$$f_{\text{bound}} = \frac{[\text{Ab}]}{K_d + [\text{Ab}]} \quad (2)$$

For 50% binding you need $[\text{Ab}] = K_d$; for 99%, $99 K_d$. So whatever Ab concentration is biologically achievable sets a *ceiling* on the K_d that is “good enough.”

6.2 K_d as a dimensionless anchor — the percentage view

Equation (2) hides a deeper point: K_d **is not a concentration, it is the scale against which antibody concentration is read**. Define $r \equiv [\text{Ab}]/K_d$. Then

$$f_{\text{bound}} = \frac{r}{1 + r},$$

a universal logistic in the single dimensionless variable r . This is the same pattern as $\text{p}K_a$, EC_{50} , or K_M . Affinity maturation is then a clean one-liner: K_d shifts left by 2–4 decades, so the same $[\text{Ab}]$ now sits 2–4 decades further right on the curve.

Where the rule of three works. When $[\text{Ab}] \ll K_d$ (the dilute regime), $K_d + [\text{Ab}] \approx K_d$ and the formula collapses to a simple proportion:

$$f_{\text{bound}} \approx \frac{[\text{Ab}]}{K_d}.$$

So at low concentrations, doubling $[\text{Ab}]$ doubles f_{bound} exactly — pure rule of three. The saturation correction (the $+[\text{Ab}]$ in the denominator) only kicks in once $[\text{Ab}]$ approaches K_d , and it prevents f_{bound} from ever exceeding 1.

6.3 What concentrations are biologically achievable?

- Total serum IgG $\sim 10 \text{ g/L} \sim 70 \mu\text{M}$ across all specificities.
- Specific Ab to one antigen post-immunisation: typically $\sim 1\text{--}100 \text{ nM}$.

- High-affinity broadly neutralising responses: up to $\sim 0.1\text{--}1\ \mu\text{M}$ specific Ab.

So specific serum antibody operates in the $10^{-9}\text{--}10^{-6}\ \text{M}$ range. For 99% occupancy you need K_d at least ~ 2 OoM below that, i.e. K_d in the $10^{-11}\text{--}10^{-8}\ \text{M}$ range.

6.4 Pathogen burden $\rightarrow$ epitope concentration in blood

$$[\text{Ep}] \approx \frac{(\text{pathogens/mL}) \cdot (\text{epitopes/pathogen})}{N_A} \cdot 10^3.$$

Bacteraemia. $\sim 10^6\ \text{CFU/mL} \times \sim 10^3\ \text{epitopes/cell} \Rightarrow [\text{Ep}] \approx 1\ \text{pM}$.

Viraemia. $\sim 10^4\text{--}10^8\ \text{virions/mL} \times \sim 10^2\ \text{trimers/virion} \Rightarrow [\text{Ep}] \approx 10^{-15}\text{--}10^{-11}\ \text{M} = \text{femto- to picomolar}$.

So target epitope concentrations in serum are sub-nanomolar to picomolar. Antibody K_d needs to sit at or below that for high-occupancy binding — exactly what affinity maturation produces. The $10^2\text{--}10^4$ affinity gain in the germinal centre is the mechanism that closes the gap between the naive K_d ($\sim 10^{-6}\ \text{M}$) and the demand ($\sim 10^{-10}\ \text{M}$).

6.5 Two worked examples: naive vs SHM-matured antibody

Same patient, same serum titer, same antigen. The only difference is that the B cell making the antibody has (or has not) undergone somatic hypermutation and selection in a germinal centre. Fix $[\text{Ab}] = 1\ \text{nM}$ for both.

Naive antibody (germline, no SHM). $K_d^{\text{naive}} = 1\ \mu\text{M} = 1000\ \text{nM}$. Plug in:

$$f_{\text{bound}}^{\text{naive}} = \frac{1\ \text{nM}}{1000\ \text{nM} + 1\ \text{nM}} = \frac{1}{1001} \approx 10^{-3} = \boxed{0.1\%}.$$

One epitope in a thousand is bound. Essentially no neutralisation.

SHM-matured antibody (after 5–7 GC cycles). $K_d^{\text{matured}} = 1\ \text{nM}$ (a thousand-fold improvement). Plug in:

$$f_{\text{bound}}^{\text{matured}} = \frac{1\ \text{nM}}{1\ \text{nM} + 1\ \text{nM}} = \frac{1}{2} = \boxed{50\%}.$$

Half the epitopes are bound. Functional neutralisation. **At the same serum titer, SHM has delivered a 500-fold gain in occupancy.**

Near the Foote–Milstein ceiling. $K_d^{\text{ceiling}} = 0.1\ \text{nM}$ (the practical kinetic ceiling). Plug in:

$$f_{\text{bound}}^{\text{ceiling}} = \frac{1\ \text{nM}}{0.1\ \text{nM} + 1\ \text{nM}} = \frac{1}{1.1} \approx \boxed{91\%}.$$

The 91% here is a property of our chosen $[\text{Ab}] = 1\ \text{nM}$. With more antibody, the same $K_d = 0.1\ \text{nM}$ would push occupancy to 99% or above. The F–M ceiling is on K_d , not on f_{bound} — it caps how *cheaply* (in serum titer) you can buy a given fraction bound.

6.6 The whole journey on one table

Antibody stage	K_d	f_{bound} at $[\text{Ab}]=1\text{ nM}$
Naive (germline)	$1\text{ }\mu\text{M}$	$\approx 0.1\%$
Partially matured	100 nM	$\approx 1\%$
Partially matured	10 nM	$\approx 9\%$
SHM-matured	1 nM	$\approx 50\%$
Near F–M ceiling	0.1 nM	$\approx 91\%$

Every order of magnitude SHM gains in K_d moves the bound fraction up one row of the table. The germinal centre’s $\sim 10^3$ improvement takes you from 0.1% bound (essentially useless) to 50% bound (functional). That is the entire reason germinal centres exist.

The flip side — antibody needed for 99% binding. Asking how much antibody it takes to reach 99% occupancy at each K_d ($[\text{Ab}] = 99K_d$) gives the sharpest argument for SHM:

Antibody stage	K_d	$[\text{Ab}]$ for 99% binding
Naive	$1\text{ }\mu\text{M}$	$99\text{ }\mu\text{M}$ — exceeds total serum IgG ($70\text{ }\mu\text{M}$)
Partial	100 nM	$9.9\text{ }\mu\text{M}$ only as dominant specificity
Partial	10 nM	990 nM feasible
SHM-matured	1 nM	99 nM easy at typical titers
Ceiling	0.1 nM	9.9 nM easy

The naive row demands more antibody than the entire IgG content of serum. **Without SHM, the serum could not physically contain enough antibody to neutralise a typical bloodborne pathogen.** Affinity maturation isn’t a luxury; it is the only way to fit the binding requirement under the production budget.

7 Closing

The repertoire is a snapshot of an ongoing process: V(D)J recombination generates a vast theoretical library, only a sample of which the animal carries, and only a sample of that has been sequenced.

The opening questions, summarised:

- Q1. About 10^6 distinct antibodies suffice for coverage of the antigenic universe (demand). The realised mouse repertoire is closer to 10^{11} – 10^{12} (sampled $\sim 10^7$, rarefaction-extrapolated higher), leaving substantial slack.
- Q2. Loss of V(D)J recombination is lethal in infancy, and jawless vertebrates independently evolved an alternative adaptive system (VLRs). Two independent inventions argue strongly for the fitness benefit.

Q3. Species differ because the V(D)J machinery admits many engineering solutions (κ -dominant, λ -dominant, ultralong CDR3, gene conversion); each lineage settles into a different point of the same design space.

And the affinity addendum from §6: a big repertoire alone is not enough — the contacts also have to be tight enough. The naive $K_d \sim 10^{-6}$ M is far from the $\sim 10^{-10}$ M demanded by realistic pathogen burdens, and the germinal centre closes that gap.

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